MAD COMBINED APPLICATION

Complete Java + XML Source Code

Registration • Login • SharedPreferences • Student Dashboard • Intents • Lifecycle • UI Components • AlertDialog • Options Menu

File	Purpose
1. RegisterActivity.java	Registration form, validation and SharedPreferences saving
2. activity_register.xml	Registration screen layout
3. MainActivity.java	Login validation and explicit navigation
4. activity_main.xml	Login screen layout
5. DashboardActivity.java	Student dashboard, widgets, storage, dialogs, menus, Intents and lifecycle
6. activity_dashboard.xml	Scrollable dashboard layout
7. main_menu.xml	Options Menu items
8. AndroidManifest.xml	Activity declarations and launcher configuration

Package name used in the examples: com.example.madcombined. If your Android Studio project uses another package name, update the package declarations and manifest consistently.

Create the menu resource directory at res/menu/ before adding main_menu.xml.

Complete MAD Lab Combined Application – Java + XML

This application combines registration, login, SharedPreferences, explicit and implicit Intents, Activity Lifecycle, CheckBox, RadioButton, Spinner, ListView, ScrollView, AlertDialog and Options Menu.

We'll use three Activities:

RegisterActivity

Create account and save registration details

MainActivity

Login and validate registered credentials

DashboardActivity

Student details, Spinner, ListView, CheckBox, RadioGroup, AlertDialog, Menu and Intents

Project configuration: Create an Android Studio project using Java and XML. Use package name `com.example.madcombined` in the examples below. If your package name is different, replace it consistently in the Java files and manifest.

1. Project structure

```
app/
 ■■■ src/main/
     ■■■ java/com/example/madcombined/
     ■   ■■■ RegisterActivity.java
     ■   ■■■ MainActivity.java
     ■   ■■■ DashboardActivity.java
     ■
     ■■■ res/
     ■   ■■■ layout/
     ■   ■   ■■■ activity_register.xml
     ■   ■   ■■■ activity_main.xml
     ■   ■   ■■■ activity_dashboard.xml
     ■   ■■■ menu/
     ■       ■■■ main_menu.xml
     ■
     ■■■ AndroidManifest.xml
```

2. RegisterActivity.java

This screen registers the user, validates the input and saves the details using SharedPreferences.

```
package com.example.madcombined;

import android.app.Activity;
import android.content.Intent;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class RegisterActivity extends Activity {

    EditText name, email, username, password, confirmPassword;
    Button register, goToLogin;
    SharedPreferences sp;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_register);

        name = findViewById(R.id.name);
        email = findViewById(R.id.email);
        username = findViewById(R.id.username);
```

```
        password = findViewById(R.id.password);
        confirmPassword = findViewById(R.id.confirmPassword);

        register = findViewById(R.id.register);
        goToLogin = findViewById(R.id.goToLogin);

        sp = getSharedPreferences("MyData", MODE_PRIVATE);

        register.setOnClickListener(v -> {
            String n = name.getText().toString().trim();
            String e = email.getText().toString().trim();
            String u = username.getText().toString().trim();
            String p = password.getText().toString();
            String cp = confirmPassword.getText().toString();

            if (n.isEmpty() || e.isEmpty() || u.isEmpty()
                    || p.isEmpty() || cp.isEmpty()) {
                Toast.makeText(this, "Please fill all fields",
                        Toast.LENGTH_SHORT).show();
            } else if (!e.contains("@")) {
                Toast.makeText(this, "Enter a valid email",
                        Toast.LENGTH_SHORT).show();
            } else if (p.length() < 4) {
                Toast.makeText(this, "Password must contain at least 4 characters",
                        Toast.LENGTH_SHORT).show();
            } else if (!p.equals(cp)) {
                Toast.makeText(this, "Passwords do not match",
                        Toast.LENGTH_SHORT).show();
            } else {
                sp.edit()
                        .putString("name", n)
                        .putString("email", e)
                        .putString("username", u)
                        .putString("password", p)
                        .putBoolean("registered", true)
                        .apply();

                Toast.makeText(this, "Registration successful",
                        Toast.LENGTH_SHORT).show();

                startActivity(new Intent(
                        RegisterActivity.this, MainActivity.class));

                finish();
            }
        });

        goToLogin.setOnClickListener(v -> {
            startActivity(new Intent(
                    RegisterActivity.this, MainActivity.class));
            finish();
        });
    }
}
```

3. activity_register.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="24dp">

        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="User Registration"
            android:textSize="26sp"
            android:textStyle="bold"
            android:gravity="center"
            android:padding="16dp" />

        <EditText
            android:id="@+id/name"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Full Name"
            android:inputType="textPersonName" />

        <EditText
            android:id="@+id/email"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
```

```
            android:hint="Email"
            android:inputType="textEmailAddress" />

        <EditText
            android:id="@+id/username"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Username"
            android:inputType="text" />

        <EditText
            android:id="@+id/password"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Password"
            android:inputType="textPassword" />

        <EditText
            android:id="@+id/confirmPassword"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Confirm Password"
            android:inputType="textPassword" />

        <Button
            android:id="@+id/register"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Register" />

        <Button
            android:id="@+id/goToLogin"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Already Registered? Login" />

    </LinearLayout>
</ScrollView>
```

4. MainActivity.java – Login

This screen checks the registered username and password from SharedPreferences. It also demonstrates an explicit Intent to open the dashboard.

```
package com.example.madcombined;

import android.app.Activity;
import android.content.Intent;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

public class MainActivity extends Activity {

    EditText username, password;
    Button login, register;
    TextView message;
    SharedPreferences sp;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        username = findViewById(R.id.username);
        password = findViewById(R.id.password);
        login = findViewById(R.id.login);
        register = findViewById(R.id.register);
        message = findViewById(R.id.message);

        sp = getSharedPreferences("MyData", MODE_PRIVATE);

        login.setOnClickListener(v -> {
            String u = username.getText().toString().trim();
            String p = password.getText().toString();

            String savedUsername = sp.getString("username", "");
            String savedPassword = sp.getString("password", "");

            if (u.isEmpty() || p.isEmpty()) {
                Toast.makeText(this, "Enter username and password",
                        Toast.LENGTH_SHORT).show();
            } else if (u.equals(savedUsername)
```

```
                    && p.equals(savedPassword)
                    && sp.getBoolean("registered", false)) {

                sp.edit().putBoolean("login", true).apply();

                Toast.makeText(this, "Login successful",
                        Toast.LENGTH_SHORT).show();

                Intent intent = new Intent(
                        MainActivity.this, DashboardActivity.class);
                startActivity(intent);
                finish();

            } else {
                Toast.makeText(this, "Invalid username or password",
                        Toast.LENGTH_SHORT).show();
            }
        });

        register.setOnClickListener(v -> {
            Intent intent = new Intent(
                    MainActivity.this, RegisterActivity.class);
            startActivity(intent);
        });

        message.setText("Please login or create a new account");
    }
}
```

5. activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Student Management Login"
        android:textSize="24sp"
        android:textStyle="bold"
        android:gravity="center"
        android:padding="16dp" />

    <EditText
        android:id="@+id/username"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Username"
        android:inputType="text" />

    <EditText
        android:id="@+id/password"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword" />

    <Button
        android:id="@+id/login"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Login" />

    <Button
        android:id="@+id/register"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Create Account" />

    <TextView
        android:id="@+id/message"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center"
        android:padding="12dp" />

</LinearLayout>
```

6. DashboardActivity.java

This is the main practical demonstration screen. It includes:

- Student details using EditText - Spinner for course selection - RadioGroup for gender - CheckBox for agreement - ListView with ArrayAdapter - SharedPreferences to save and retrieve student details - AlertDialog - Explicit and implicit Intents - Options Menu - Activity Lifecycle callbacks

```
package com.example.madcombined;

import android.app.Activity;
import android.app.AlertDialog;
import android.content.Intent;
import android.content.SharedPreferences;
import android.net.Uri;
import android.os.Bundle;
import android.util.Log;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.EditText;
import android.widget.ListView;
import android.widget.RadioGroup;
import android.widget.Spinner;
import android.widget.TextView;
import android.widget.Toast;

public class DashboardActivity extends Activity {

    EditText studentName, rollNumber;
    Spinner courseSpinner;
    RadioGroup genderGroup;
    CheckBox agree;
    ListView listView;
    TextView welcome;
    Button save, load, reset, alert, dial, share, logout;

    SharedPreferences sp;
    ArrayAdapter<String> adapter;

    String[] subjects = {
            "Java", "Android", "Python", "Database", "Computer Networks"
    };

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_dashboard);

        Log.d("LIFECYCLE", "onCreate");

        studentName = findViewById(R.id.studentName);
        rollNumber = findViewById(R.id.rollNumber);
        courseSpinner = findViewById(R.id.courseSpinner);
        genderGroup = findViewById(R.id.genderGroup);
        agree = findViewById(R.id.agree);
        listView = findViewById(R.id.listView);
        welcome = findViewById(R.id.welcome);

        save = findViewById(R.id.save);
        load = findViewById(R.id.load);
        reset = findViewById(R.id.reset);
        alert = findViewById(R.id.alert);
        dial = findViewById(R.id.dial);
        share = findViewById(R.id.share);
        logout = findViewById(R.id.logout);

        sp = getSharedPreferences("MyData", MODE_PRIVATE);

        String user = sp.getString("name", "Student");
        welcome.setText("Welcome, " + user);

        // Spinner
        String[] courses = {"MCA", "MCS", "BCA", "BSc Computer Science"};

        ArrayAdapter<String> courseAdapter =
                new ArrayAdapter<>(this,
                        android.R.layout.simple_spinner_item, courses);

        courseAdapter.setDropDownViewResource(
                android.R.layout.simple_spinner_dropdown_item);

        courseSpinner.setAdapter(courseAdapter);

        // ListView with ArrayAdapter
        adapter = new ArrayAdapter<>(this,
                android.R.layout.simple_list_item_1, subjects);
```

```
listView.setAdapter(adapter);

listView.setOnItemClickListener((parent, view, position, id) -> {
    Toast.makeText(this, "Selected: " + subjects[position],
            Toast.LENGTH_SHORT).show();
});

// Save student details
save.setOnClickListener(v -> {
    String name = studentName.getText().toString().trim();
    String roll = rollNumber.getText().toString().trim();

    if (name.isEmpty() || roll.isEmpty()) {
        Toast.makeText(this, "Enter all student details",
                Toast.LENGTH_SHORT).show();
        return;
    }

    if (!agree.isChecked()) {
        Toast.makeText(this, "Please accept the agreement",
                Toast.LENGTH_SHORT).show();
        return;
    }

    int selectedId = genderGroup.getCheckedRadioButtonId();

    if (selectedId == -1) {
        Toast.makeText(this, "Select gender",
                Toast.LENGTH_SHORT).show();
        return;
    }

    String gender = selectedId == R.id.male
            ? "Male" : "Female";

    String course = courseSpinner.getSelectedItem().toString();

    sp.edit()
            .putString("studentName", name)
            .putString("rollNumber", roll)
            .putString("gender", gender)
            .putString("course", course)
            .apply();

    Toast.makeText(this, "Student details saved",
            Toast.LENGTH_SHORT).show();
});

// Load saved student details
load.setOnClickListener(v -> {
    studentName.setText(sp.getString("studentName", ""));
    rollNumber.setText(sp.getString("rollNumber", ""));

    String savedCourse = sp.getString("course", "MCA");
    int position = courseAdapter.getPosition(savedCourse);

    if (position >= 0) {
        courseSpinner.setSelection(position);
    }

    String gender = sp.getString("gender", "");

    if (gender.equals("Male")) {
        genderGroup.check(R.id.male);
    } else if (gender.equals("Female")) {
        genderGroup.check(R.id.female);
    }

    Toast.makeText(this, "Student details loaded",
            Toast.LENGTH_SHORT).show();
});

// Reset student details
reset.setOnClickListener(v -> {
    studentName.setText("");
    rollNumber.setText("");
    genderGroup.clearCheck();
    agree.setChecked(false);
    courseSpinner.setSelection(0);

    sp.edit()
            .remove("studentName")
            .remove("rollNumber")
            .remove("gender")
            .remove("course")
            .apply();

    Toast.makeText(this, "Student details reset",
            Toast.LENGTH_SHORT).show();
});
```

```
    // AlertDialog
    alert.setOnClickListener(v -> {
        new AlertDialog.Builder(DashboardActivity.this)
                .setTitle("Confirmation")
                .setMessage("Do you want to continue?")
                .setPositiveButton("Yes", (dialog, which) -> {
                    Toast.makeText(this, "You selected Yes",
                            Toast.LENGTH_SHORT).show();
                })
                .setNegativeButton("No", (dialog, which) -> {
                    dialog.dismiss();
                })
                .show();
    });

    // Implicit Intent: open phone dialler
    dial.setOnClickListener(v -> {
        Intent intent = new Intent(Intent.ACTION_DIAL);
        intent.setData(Uri.parse("tel:9876543210"));
        startActivity(intent);
    });

    // Implicit Intent: share text
    share.setOnClickListener(v -> {
        Intent intent = new Intent(Intent.ACTION_SEND);
        intent.setType("text/plain");
        intent.putExtra(Intent.EXTRA_TEXT,
                "Hello! This is a Student Management Application.");

        startActivity(Intent.createChooser(intent, "Share using"));
    });

    // Logout
    logout.setOnClickListener(v -> {
        sp.edit().putBoolean("login", false).apply();

        Intent intent = new Intent(
                DashboardActivity.this, MainActivity.class);

        startActivity(intent);
        finish();
    });
}

// Options Menu
@Override
public boolean onCreateOptionsMenu(Menu menu) {
    getMenuInflater().inflate(R.menu.main_menu, menu);
    return true;
}

@Override
public boolean onOptionsItemSelected(MenuItem item) {

    int id = item.getItemId();

    if (id == R.id.menuHome) {
        Toast.makeText(this, "Home selected",
                Toast.LENGTH_SHORT).show();
        return true;

    } else if (id == R.id.menuAbout) {
        new AlertDialog.Builder(this)
                .setTitle("About")
                .setMessage("Student Management Application\nMAD Lab")
                .setPositiveButton("OK", null)
                .show();
        return true;

    } else if (id == R.id.menuLogout) {
        sp.edit().putBoolean("login", false).apply();

        startActivity(new Intent(
                DashboardActivity.this, MainActivity.class));
        finish();
        return true;
    }

    return super.onOptionsItemSelected(item);
}

// Activity Lifecycle
@Override
protected void onStart() {
    super.onStart();
    Log.d("LIFECYCLE", "onStart");
}

@Override
protected void onResume() {
    super.onResume();
```

```
        Log.d("LIFECYCLE", "onResume");
    }

    @Override
    protected void onPause() {
        super.onPause();
        Log.d("LIFECYCLE", "onPause");
    }

    @Override
    protected void onStop() {
        super.onStop();
        Log.d("LIFECYCLE", "onStop");
    }

    @Override
    protected void onRestart() {
        super.onRestart();
        Log.d("LIFECYCLE", "onRestart");
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        Log.d("LIFECYCLE", "onDestroy");
    }
}
```

7. activity_dashboard.xml

The ScrollView allows the user to scroll through the complete dashboard.

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">

        <TextView
            android:id="@+id/welcome"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Welcome"
            android:textSize="24sp"
            android:textStyle="bold"
            android:padding="12dp" />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Student Details"
            android:textSize="20sp"
            android:textStyle="bold" />

        <EditText
            android:id="@+id/studentName"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Student Name"
            android:inputType="textPersonName" />

        <EditText
            android:id="@+id/rollNumber"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Roll Number"
            android:inputType="number" />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Select Course"
            android:textStyle="bold" />

        <Spinner
            android:id="@+id/courseSpinner"
            android:layout_width="match_parent"
            android:layout_height="wrap_content" />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
```

```
            android:text="Gender"
            android:textStyle="bold"
            android:paddingTop="12dp" />

        <RadioGroup
            android:id="@+id/genderGroup"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="horizontal">

            <RadioButton
                android:id="@+id/male"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Male" />

            <RadioButton
                android:id="@+id/female"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Female" />

        </RadioGroup>

        <CheckBox
            android:id="@+id/agree"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="I agree to the terms and conditions" />

        <Button
            android:id="@+id/save"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Save Student Details" />

        <Button
            android:id="@+id/load"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Load Student Details" />

        <Button
            android:id="@+id/reset"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Reset Student Details" />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Subject List"
            android:textSize="20sp"
            android:textStyle="bold"
            android:paddingTop="16dp" />

        <ListView
            android:id="@+id/listView"
            android:layout_width="match_parent"
            android:layout_height="200dp"
            android:nestedScrollingEnabled="false" />

        <Button
            android:id="@+id/alert"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Show Alert Dialog" />

        <Button
            android:id="@+id/dial"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Open Dialler" />

        <Button
            android:id="@+id/share"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Share Message" />

        <Button
            android:id="@+id/logout"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Logout" />

    </LinearLayout>
</ScrollView>
```

8. Options Menu – main_menu.xml

Create a `menu` resource directory under `res` if it does not already exist. Then create `main_menu.xml` inside it.

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">

    <item
        android:id="@+id/menuHome"
        android:title="Home" />

    <item
        android:id="@+id/menuAbout"
        android:title="About" />

    <item
        android:id="@+id/menuLogout"
        android:title="Logout" />

</menu>
```

9. AndroidManifest.xml

Register all three Activities. MainActivity is the launcher Activity.

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <application
        android:allowBackup="true"
        android:label="MAD Combined App"
        android:theme="@android:style/Theme.Material.Light.DarkActionBar"
        android:supportsRtl="true">

        <activity
            android:name=".DashboardActivity"
            android:exported="false" />

        <activity
            android:name=".RegisterActivity"
            android:exported="false" />

        <activity
            android:name=".MainActivity"
            android:exported="true">

            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>

        </activity>

    </application>

</manifest>
```